COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithm	**SECTION:**	CPE21S2
DATE PERFORMED:	09/18/2026	**DATE SUBMITTED:**	09/18/2026
NAME:	Milan, Kherwin Emmanuel Moradillo, Eunice Nieles, Cenndy Realubit, Seth Andrew	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

ACTIVITY NO. 9.1 Binary Search Tree

1. PROCEDURE OUTPUT

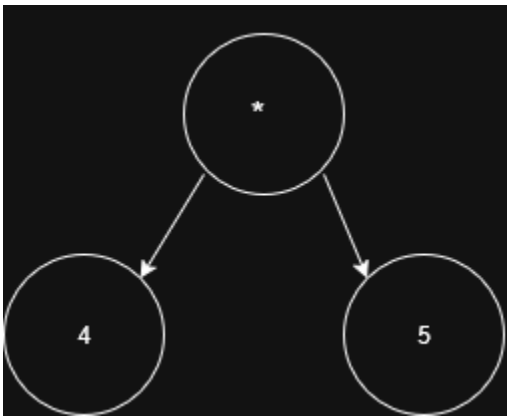
What procedures did you perform?

1. What is a binary search tree?
   - A **Binary Search Tree (BST)** is a type of binary data structure where each node stores a value and keys, which these nodes can have two smaller nodes below it called children. BST satisfies a specific property, All nodes in the left subtree of a node contain values strictly less than the node's value. All nodes in the right subtree of a node contain values strictly greater than the node's value. For a balanced binary search tree, search, insertion, and deletion all run in O(log n) time. But in the worst case, when the tree is unbalanced, these operations slow down to O(n).

2. Where can binary search trees be used?
   - Binary search trees are useful for finding things quickly and keeping data in order. They're used for fast lookup, add, and removal of key-value pairs, in databases (often as B-trees) to find records quickly, in autocomplete and spell-checkers to match words fast, and in some file systems to organize folders. They also help with priority scheduling, sorting (an in-order walk gives sorted data), and finding values between two numbers, since the tree is already ordered. Some networking systems use trees for fast routing lookups, and trees show up in parsing too. In practice, plain BSTs are usually replaced by balanced versions like AVL, Red-Black trees, or B-trees, since these stay fast even in the worst case.

3. What is a tree traversal?
   - A tree traversal refers to the process of accessing each node in a tree in a specific order. Tree traversals offer two categories, the Depth-First Traversal (DFT) and Breadth-First Traversal (BFT). The DFT explores as far as possible along a branch before exploring the next branch. It has 3 types: In-order (left, node, right), Pre-order (node, left, right), Post-order (left, right, node). The BFT explores nodes level by level from top to bottom. It has one type: Level-order that visits all nodes level by level, top to bottom, left to right within each level. This uses a queue rather than recursion.

4. Differentiate a post-order and pre-order traversal, give examples.

Pre-Order	Post-Order
• Follows the rule of root -> left -> right • Works by starting from the top to the bottom	• Follows the rule of left -> right -> root • Works by starting from the bottom to the top
EXAMPLE:	
 Pre-order reads the diagram as [* 4 5] meanwhile, post-order will read it as [4 5 *], from the computer's perspective the pre-order would look as multiplying these two numbers while the post-order would read the 4 and 5 first before recognizing that it should be multiplied.	

5. What is a parse tree and its use?

   A parse tree or also known as concrete syntax tree, is a hierarchical, tree-like data structure that represents something similar to a family tree for a sentence or a line of syntax. Instead of showing the parent and children, it shows how words or symbols fit together.

   It is mainly used for compilers and grammar checkers. For compilers, the computer builds a tree to make sure you didn't make a typo and figure out mathematical equations. Meanwhile, apps/websites like Grammarly and alike use this for spotting if a word is in the wrong place.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic

work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.